Bubbles and Beliefs:
Three Plans for Utility-Agent Belief Update

Wang*

April 2026

Abstract. We replicate the finite-horizon experimental asset market of Dufwenberg et al. (2005) inside a deterministic browser-based simulator and embed it with the expected-utility agent of Lopez-Lira (2025). Holding the market substrate, the agent population, the prior, and the action choice rule fixed, we then compare three distinct belief-update *plans* for the utility trader's per-period subjective valuation. *Plan I* is an offline algorithmic blend of the agent's own prior with the mean foreign peer claim, weighted by an endogenous experience counter k_i that ramps the weight from 0.60 to 0.90 across the four-round session. *Plan II* replaces the blend with a period-boundary chat-completion call whose prompt gives the language model the closed-form risk-utility expression $U \in \{U_L, U_N, U_A\}$ for that agent's risk type. *Plan III* uses the same wiring as Plan II but strips the functional forms out of the prompt and only names the risk-preference category. All three plans feed their output V_i^{post} into the identical Lopez-Lira expected-utility scoring step, so any difference in the resulting market dynamics is attributable to the update channel alone. The simulator reports Haessel R^2, normalised deviation, amplitude, turnover, and allocative efficiency for all three plans; the seeded mulberry32 PRNG makes within-plan replays exact, and Plan II and Plan III silently degrade to Plan I for any agent whose LLM call fails, so every run can be reproduced without network access. This paper documents the replication, the three update plans, and the market-quality instruments used to compare them.

Keywords: experimental asset markets, bubbles, expected utility, language models, belief updating, allocative efficiency.

JEL Codes: C92, D84, G12, G41.

I. Introduction

Experimental asset markets in the tradition of Smith et al. (1988) have produced one of the most replicated departures from rational-expectations pricing in all of economics: even when subjects know, in closed form, that the fundamental value of a finite-lived asset declines deterministically with the remaining dividend stream, transaction prices systematically overshoot the fundamental before collapsing near the terminal period. Dufwenberg et al. (2005)—hereafter DLM 2005—established a by now canonical result in this literature: when two-thirds or more of the subject pool consists of traders who have played the game before, the bubble disappears. Experience, rather than cognition or incentives, is what eventually kills the bubble.

A parallel and much more recent literature replaces the human subjects with AI agents. Lopez-Lira (2025) proposes an expected-utility framework in which each AI-agent trader carries a persistent valuation bias b_i, a risk preference over wealth $U \in \{U_L, U_N, U_A\}$, and a belief-blending channel that lets agents exchange private valuations and update a trust score for each sender by exponential moving average. Lopez-Lira's agents optimise expected utility over a small action set—hold, cross the bid, cross the ask, post passively—and the resulting population can be dialled between skeptical and naive, truthful and deceptive, risk-averse and risk-loving by simple categorical knobs. The Lopez-Lira design is a natural vehicle for studying how a heterogeneous AI-agent population prices a DLM asset, but it leaves one channel unspecified: *how should*

*Computational replication and source code are available at `https://github.com/Mon-ius/bubble`. The author thanks the maintainers of the original DLM and Lopez-Lira testbeds for making their designs easy to re-implement. All errors are the author's own.

each utility agent translate its prior and the messages it has received into a posterior valuation that it then carries into the next period? Every downstream instrument—the expected-utility score, the order chosen, the trust EMA, the order-matching step, the dividend credit—consumes that posterior V_i^{post} and is otherwise unaffected by the belief-update channel.

This paper contributes a computational testbed that holds the market substrate, the agent population, the starting prior, and the action-choice rule strictly fixed, and compares three distinct *plans* for the per-period belief update.

Plan I (algorithmic baseline).. The posterior is a convex combination of the agent's own prior and the arithmetic mean of foreign peer claims received during the period, with a weight w that ramps endogenously with experience,

$$V_i^{\text{post}} \;=\; w \cdot V_i^{\text{prior}} \;+\; (1-w) \cdot \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \hat{v}_m, \qquad w = 0.6 + 0.1\,\min(3,\, k_i), \tag{1}$$

where k_i is the agent's `roundsPlayed` counter. A novice ($k_i = 0$) moves appreciably toward peers ($w = 0.60$), a three-round veteran ($k_i \geq 3$) barely moves ($w = 0.90$). No network calls are placed; the update is deterministic under the seeded PRNG.

Plan II (LLM update with explicit utility forms).. At every period boundary the engine fires one parallel `chat/completions` call per utility agent. The user prompt encodes the market constants, the agent's risk preference and bias mode, the peer claims accumulated during the period, the agent's mark-to-market wealth, and—critically—the *closed-form* expression of the agent's risk-utility functional:

$$\pi_i^{\text{II}} \;\supset\; \left\{ U_L(w) = (w/w_0)^2,\; U_N(w) = w/w_0,\; U_A(w) = \sqrt{w/w_0} \right\}. \tag{2}$$

The model is asked to return a single number in experimental cents, which the engine parses, clamps to a sanity interval, and writes to the agent's next-period prior slot. The call is fire-and-forget: if the network errors or the parse fails, the agent silently degrades to Plan I for that period, so reproducibility is preserved even when the key is absent.

Plan III (LLM update with risk label only).. Identical wiring to Plan II, but the user prompt strips the closed-form expressions out of equation (2) and names only the risk-preference category:

$$\pi_i^{\text{III}} \;\supset\; \{\text{risk-loving} \,|\, \text{risk-neutral} \,|\, \text{risk-averse}\}. \tag{3}$$

Plan III therefore isolates a narrow, purely linguistic question: does handing the model an explicit functional form for utility produce different market dynamics than naming the risk type alone? The LLM in Plan III must translate "risk-averse" into an implicit curvature internally; the LLM in Plan II is handed the curvature as a formula. Any divergence between the two plans' market statistics quantifies the prompt-sensitivity of the model's internal expected-utility reasoning.

Research question.. With the market substrate, the agent population, the Lopez-Lira prior, the expected-utility action-choice rule, the order-matching engine, and the seed all held fixed, the question the testbed is built to answer is

$$\text{Plan I} \;\overset{?}{\approx}\; \text{Plan II} \;\overset{?}{\approx}\; \text{Plan III} \quad \text{in classical bubble statistics and posterior trajectory?}$$

A null result (all three plans produce indistinguishable market dynamics) would imply that the specific belief-update channel is absorbed by the downstream expected-utility score and does not propagate into observable market behaviour. A divergence between Plan I and the two LLM plans would implicate the LLM itself as a source of systematic behavioural difference. A divergence *between* Plan II and Plan III would implicate prompt structure—specifically, whether handing the model a closed-form utility function alters its valuation choice—as a source of behavioural difference that goes beyond the agent's declared risk type.

Section II places the simulator in the literature on experimental asset markets, expected-utility AI-agent traders, and LLM prompt sensitivity. Section III documents the market structure and the sampling stage. Section IV gives the precise decision rule for each agent type. Section V presents the three belief-update plans: the algorithmic blend, the two LLM calls, the prompt templates, the clamp, the consume-and-cache

path, and the fallback-to-Plan I invariant. Section VI reports preliminary findings, and Section VII concludes.

II. Related Literature

The simulator sits at the intersection of four strands of literature: the experimental-bubble tradition, the heterogeneous agent-based finance tradition, the expected-utility decision-theoretic tradition, and the emerging work on AI agents as economic subjects.

Experimental bubbles.. The taxonomy of the market in this paper is directly inherited from Smith et al. (1988), who introduced the declining-fundamental, finite-horizon asset and documented the characteristic bubble-and-crash trajectory in a continuous double auction. Dufwenberg et al. (2005) show that the Smith–Suchanek–Williams bubble is fragile to *experience* rather than to instructions, incentives, or trading institutions: once two-thirds of the subjects have played the game before, transaction prices track the fundamental staircase from the first period onward. The testbed in this paper inherits the DLM 2005 market substrate—asset life, dividend distribution, fundamental staircase, four-round session loop—without modification, and reuses the endogenous experience counter k_i to ramp Plan I's belief-blend weight across the session, mirroring the paper's procedural-experience channel at the level of the agent's belief update rather than the agent's decision rule.

Heterogeneous agent-based finance.. Three classical rules supply the pinned background of the simulator: a *Fundamentalist* (F) that mean-reverts toward FV_t in the Smith–Suchanek–Williams lineage; a *Trend follower* (T) that implements the positive-feedback strategy of De Long et al. (1990) by extrapolating a six-tick slope; and a *Random* (R) zero-intelligence trader after Gode and Sunder (1993) that posts uniform bids on $[0.7, 1.1]\,\mathrm{FV}_t$ and uniform asks on $[0.9, 1.3]\,\mathrm{FV}_t$. Together these three rules provide a rational anchor, a momentum amplifier, and a liquidity floor; the utility agent layers an expected-utility optimiser on top and is the class whose belief-update plan the present paper varies.

Expected utility and AI-agent traders.. Lopez-Lira (2025) introduces a utility-maximising AI-agent trader whose prior is a perturbation of FV_t, whose action space is $\{\text{hold}, \text{cross bid}, \text{cross ask}, \text{passive bid}, \text{passive ask}\}$, and whose choice rule picks the action with the highest expected utility, $\mathrm{EU}(a) = p_{\text{fill}} \cdot U(w_1) + (1 - p_{\text{fill}}) \cdot U(w_0)$. Agents carry one of three risk-preference functionals $\{U_L(w), U_N(w), U_A(w)\} = \{(w/w_0)^2,\ w/w_0,\ \sqrt{w/w_0}\}$ and exchange private valuations through a lightweight messaging bus. A pairwise trust score $\tau_{r\to s} \in [0, 1]$ is updated by exponential moving average whenever a claimed valuation is observed against the period's volume-weighted average trade price. The present paper inherits this framework exactly as described—the prior, the action set, the EU score, the risk functionals, the trust EMA, and the messaging bus are all reimplemented without modification—and then varies only the period-boundary belief update by comparing the algorithmic blend of Plan I against the language-model calls of Plan II and III.

Language models in economic decision tasks.. A small but rapidly growing literature studies whether language models reproduce well-documented behavioural regularities when embedded in economic decision tasks, and whether their outputs are sensitive to prompt structure. The contrast between Plan II and Plan III is a specific instance of this broader question: Plan II hands the model the closed-form expression of the risk-utility functional U_L, U_N, U_A; Plan III hands the model only the risk label. Both prompts route through the same clamp, the same consume-and-cache pipeline, and the same expected-utility scoring rule downstream, so any observed difference in the posterior distribution or in downstream market statistics is attributable to the prompt structure alone. This narrow A/B design makes prompt-sensitivity measurable on the same scale as the canonical bubble statistics.

Market-quality instruments.. The testbed reports four standard bubble statistics—Haessel's R^2 (Haessel, 1978), normalised absolute deviation, amplitude, and turnover, following SSW and Stöckl et al. (2010)—alongside allocative efficiency,

$$\mathrm{AE} \;=\; \frac{\sum_i V_i^{\text{post}} \cdot \text{inventory}_i}{\max_i V_i^{\text{post}} \cdot Q},$$

which quantifies the fraction of theoretically-maximal utility realised at the end of the session given the final posterior valuations. All five instruments are reported for every plan on every run, so cross-plan comparisons can be made without any extra code.

III. Experimental Design

A. Session structure: four consecutive rounds

The unit of study in Dufwenberg et al. (2005) is a *session* of four consecutive ten-period markets played by the same agent population. The paper insists on the terminology explicitly:

> "A session involved four consecutive markets. In the following, we shall talk in terms of four different rounds. Note the distinction between rounds and periods; a round (being a market) consists of ten periods." (Dufwenberg et al., 2005, p. 1733)

The session wrapper is the load-bearing element of the original design: in the human-subject study, bubbles inflate in round 1, attenuate in rounds 2–3, and have effectively died by round 4 even though the population is identical—experience, not composition, kills the bubble. The testbed inherits this wrapper because it is also the channel through which Plan I's experience-weighted trust runs: every surviving utility agent's `roundsPlayed` counter is incremented at the end of each non-final round, which feeds equation (13) and ramps the Plan I belief-blend weight from 0.60 to 0.90 across the session. The simulator therefore wraps the per-round market in a session loop of $R = 4$ rounds. At every round boundary the order book is cleared, each agent's cash and inventory are rewound to its sampled spec, and the period counter resets to one. Learned state—trust matrices, the experience counter, the candidate action cache—is intentionally preserved across the boundary so that the cross-round learning channel that drives Plan I's weight ramp is the only source of persistent change between consecutive markets. Both R and T are surfaced read-only in the simulator's "Paper constants" panel and are not user tunable.

B. The asset and the fundamental

The asset is a finite-lived claim on an iid per-period dividend that pays either zero or twenty experimental cents with equal probability. Letting d denote the period-t dividend draw,

$$d \sim \text{Uniform}\{0, 20\}, \qquad \mathbb{E}[d] = 10, \qquad T = 10. \tag{4}$$

The asset lives for $T = 10$ periods within each round and pays nothing thereafter, so its risk-neutral fundamental value at the start of period t of any round is

$$\text{FV}_t \;=\; \mathbb{E}[d] \cdot (T - t + 1) \;=\; 10 \cdot (T - t + 1), \tag{5}$$

that is, a linearly declining staircase from $\text{FV}_1 = 100$ to $\text{FV}_T = 10$ with a terminal value of zero after the final dividend pays. The fundamental resets to $\text{FV}_1 = 100$ at the start of every new round, so the price chart traces a four-tooth saw-tooth across the full session. Equation (4) and equation (5) reproduce the dividend process and the fundamental schedule of Dufwenberg et al. (2005) exactly. The session length R, the per-round asset life T, and the dividend mean $\mathbb{E}[d]$ are all surfaced read-only in the simulator's "Paper constants" panel so that every comparison across plans holds the market structure fixed.

C. Continuous double auction with period discretisation

The simulator discretises each period into $K = 18$ decision rounds, so a single round comprises $T \cdot K = 180$ ticks and a full four-round session comprises $R \cdot T \cdot K = 720$ ticks. Every tick, each agent posts at most one order, routed to a price–time priority order book; a crossing pair matches at the resting price. At the end of each period the outstanding quotes are withdrawn, the dividend is drawn from (4) and credited to each holder, and the belief-update plan (Plan I, II, or III) runs once to write next period's V_i^{prior}. At the end of period T of every non-final round the session loop also rewinds endowments, clears the book, increments every surviving agent's `roundsPlayed` counter, and resets the period counter to one before the next round begins. The $K = 18$ discretisation is the simulator's own approximation of the two-minute continuous z-Tree window used in the paper, is pinned in the "Simulator constants" panel, and is not user tunable.

D. Population and endowment sampling

The population is a mix of fundamentalists (F), trend followers (T), zero-intelligence randoms (R), and utility agents (U), dialled from the Experiment Settings panel. A utility slot is always present because Plans I, II, and III act only on U agents—the background slots (F, T, R) are there to provide a rational anchor, a momentum amplifier, and a liquidity floor, and play the same role under every plan.

Before the engine starts, `sampleAgents(mix, rng)` draws a flat list of agent specs, each carrying a numeric id, an agent type label, a personal name drawn without replacement from a built-in pool, a cash endowment drawn uniformly from $[800, 1200]$ cents, and an inventory drawn uniformly from $\{2, 3, 4\}$. Utility agents additionally carry a persistent bias mode b_i and risk-preference type U_i, assigned from a user-controlled split over optimistic/unbiased/pessimistic and loving/neutral/averse. The sample RNG is derived from $\text{seed} \oplus$ `0xA5A5A5A5` and is intentionally independent of the engine RNG, so editing a cash or inventory value in the Agents panel and pressing Rebuild produces the same per-tick trading sequence as a matched run without the edit.

E. Reproducibility

Every run is driven by a single scalar seed. A mulberry32 generator $s_{k+1} = (s_k + \texttt{0x6D2B79F5}) \bmod 2^{32}$ produces the $(0, 1)$-uniform stream consumed by every stochastic step—dividend realisations, zero-intelligence quotes, the noise term in the utility prior, and the trust-update ordering. Calls to `Math.random` from agent or market code are a contract violation that breaks reproducibility by the (population, seed) pair and are flagged as an invariant in the project style guide. The seed is re-rolled only on Reset, and the replay system reconstructs past states by slicing append-only history arrays on `Market` and `Logger`—no frame is ever recomputed. A single seed therefore reproduces the entire 720-tick session end-to-end, rounds and round-end resets included.

Plans II and III add a single exogenous source of randomness—the language-model reply—which the engine handles by keeping the Plan I fallback closed-form (§C, §D). Paired comparisons across plans therefore always carry a (plan, seed) tag, and a Plan II run with zero successful LLM calls is definitionally identical to the matching Plan I run.

IV. Agent Model

This section gives the exact decision rule for each of the four agent classes in the simulator. All rules read the top of the book (best bid b_t and best ask a_t), the current fundamental value FV_t, and a seeded pseudo-random draw $r \sim \text{Uniform}(0, 1)$; the formulas below are the literal rules implemented in the source code, with constants exposed to the reader so that any experiment can be reconstructed from this page alone. The utility agent (§D) is the only class whose period-boundary belief update differs across the three plans of Section V; every other class runs an identical rule under all three plans and is described once here.

A. Fundamentalist (F)

The fundamentalist is a rational mean-reversion anchor inspired by Smith et al. (1988). Its belief is $\beta_t = \mathrm{FV}_t \cdot \big(1 + 0.02\,(r - 0.5) \cdot 2\big)$—a $\pm 2\%$ symmetric noise about the fundamental—and its action rule is

$$\begin{cases} \text{cross the ask with a unit bid at } a_t, & \text{if } a_t < 0.98\,\beta_t \text{ and cash} \geq a_t, \\ \text{cross the bid with a unit ask at } b_t, & \text{if } b_t > 1.02\,\beta_t \text{ and inventory} > 0, \\ \text{post passive with probability } 0.45, & \text{otherwise}, \\ \text{hold}, & \text{otherwise}. \end{cases} \tag{6}$$

Passive quotes bid in $[0.970,\ 1.005] \cdot \beta_t$ and ask in $[0.995,\ 1.030] \cdot \beta_t$, so the fundamentalist never posts a cross on its own initiative. It supplies a narrow two-sided book near the fundamental whenever neither side of the market is aggressive and anchors mispricing back to FV_t through unit crosses when either side drifts by more than 2%.

B. Trend follower (T)

The trend follower estimates a short-horizon slope from the last six non-null trade prices and projects three ticks forward,

$$\text{slope}_t \;=\; \frac{p_t - p_{t-h+1}}{\max(1, h-1)}, \qquad \text{target}_t \;=\; p_t + 3 \cdot \text{slope}_t, \tag{7}$$

where $h = 6$ is the slope window and p_t is the most recent trade price. When $\text{slope}_t > 0$ and it has cash, the trend agent chases up with a unit bid at $p_t \cdot (1 + 0.015 + 0.03\,r)$; when $\text{slope}_t < 0$ and it has inventory, it flees with a unit ask at $p_t \cdot (1 - 0.015 - 0.03\,r)$. Two corner cases protect the agent from stranding. First, if cash has dropped below the most recent trade price but inventory still exceeds the initial endowment, the agent dumps shares into the best bid at the aggressive price. Second, in a flat-slope regime with excess inventory, the agent liquidates with probability $\min(0.85,\ 0.3 + 0.25\,e)$, where e is the excess-inventory rate. The trend rule is the positive-feedback channel (De Long et al., 1990) and is the primary amplifier of bubble formation in the simulator.

C. Random zero-intelligence (R)

The zero-intelligence trader posts uniform bids and asks on fundamentally-scaled support, following Gode and Sunder (1993). Explicitly,

$$\text{bid}_t \sim \text{Uniform}[\,0.7\,\text{FV}_t,\ 1.1\,\text{FV}_t\,], \qquad \text{ask}_t \sim \text{Uniform}[\,0.9\,\text{FV}_t,\ 1.3\,\text{FV}_t\,]. \tag{8}$$

It never reads the order book, never tracks profit, and is pinned in every run so that the book never goes fully empty. One slot of this type is part of the fixed background and is invisible to the population-composition sliders.

D. Utility (U)

The utility agent follows Lopez-Lira (2025). On every decision tick its prior is

$$V_i^{\text{prior}} \;=\; \text{FV}_t \cdot \big(1 + b_i + \varepsilon_{i,t}\big), \qquad b_i \in \{-0.15,\ 0,\ +0.15\}, \qquad \varepsilon_{i,t} \sim \text{Uniform}[-0.03,\ 0.03]. \tag{9}$$

The persistent bias b_i is the "persistently optimistic/pessimistic/unbiased" archetype and is assigned at sampling time. The per-tick noise ε keeps the agent's quoting out of lockstep with other agents of the same type.

During a period, utility agents exchange private valuations through a lightweight messaging bus. Each agent accumulates a `receivedMsgs` array of peer claims $\{\hat{v}_m\}$ over the course of the period. At the period boundary the engine invokes the active belief-update plan (Plan I, II, or III; see Section V), which consumes the prior, the message set, and the agent's mark-to-market wealth, and returns the period-end posterior V_i^{post}. That posterior becomes the next period's V_i^{prior}.

The trust bookkeeping is unchanged across the three plans. Pairwise trust is updated every time a claimed valuation is observed against the period's volume-weighted average trade price:

$$\tau_{r\to s} \;\leftarrow\; (1-\alpha)\,\tau_{r\to s} \;+\; \alpha \cdot \max\!\Big(0,\ 1 - \tfrac{|\hat{v}_s - \text{VWAP}_t|}{\text{VWAP}_t}\Big), \qquad \alpha = 0.30. \tag{10}$$

Receiver r's trust in sender s is reinforced whenever s's claim is close to the period VWAP and attenuated otherwise. Plan I uses $\tau_{r\to s}$ only indirectly through the experience-weighted blend (Section B); Plans II and III can surface it in the prompt as a messaging-context variable (Section C).

Given the posterior valuation V_i^{post}, the agent scores each available action $a \in \{\text{hold, cross bid, cross ask, passive bid, pass}$ by its expected utility

$$\text{EU}(a) \;=\; p_{\text{fill}} \cdot U(w_1) \;+\; (1 - p_{\text{fill}}) \cdot U(w_0), \qquad a^\star = \arg\max_a \text{EU}(a), \tag{11}$$

where w_0 and w_1 are the agent's wealth under non-fill and fill respectively, and $p_{\text{fill}} = 0.30$ is a constant passive fill-probability inherited from the original Lopez-Lira design. The risk-preference functional $U \in \{U_L, U_N, U_A\}$ is assigned at sampling time and drawn from

$$U_L(w) = (w/w_0)^2, \qquad U_N(w) = w/w_0, \qquad U_A(w) = \sqrt{w/w_0}. \tag{12}$$

Equations (6)–(12) constitute the decision step that is shared across all three belief-update plans. The market aggregates the resulting orders through a price–time-priority order book, mutates cash and inventory

on every trade, credits dividends per (4) at period end, and records every state in append-only history arrays so the replay system can reconstruct past ticks without re-running the agents. The next section specifies Plan I, Plan II, and Plan III in full.

V. Three Belief-Update Plans

A. Overview

This section specifies the three *plans* by which a utility agent translates its prior and the messages it has received into a posterior valuation each period. Every agent starts with the same Lopez-Lira prior of equation (9); every agent scores candidate orders with the same expected-utility rule of equation (11); every plan feeds its output—a single number V_i^{post}, in experimental cents—into that score. The only thing that differs across plans is the function that maps the prior plus peer messages into the posterior.

Plan I is deterministic and runs closed-form. Plan II and Plan III fire a parallel LLM call per utility agent at every period boundary, and differ only in the prompt: Plan II hands the model the closed-form risk-utility expression for that agent's type, while Plan III names only the risk-preference category. A failed call in Plan II or Plan III degrades silently to Plan I for that agent in that period, so every run is replayable even when the key is absent and the mulberry32 PRNG alone drives the downstream dynamics.

B. Plan I: algorithmic blend with experience-weighted trust

Let V_i^{prior} denote the agent's Lopez-Lira prior from equation (9), let $\mathcal{M}_i$ be the set of claimed valuations received from foreign peers during the current period (messages from the agent itself are excluded), and let $k_i = \texttt{roundsPlayed}_i \in \{0, 1, 2, \ldots\}$ be the agent's endogenous experience counter. Plan I computes the period-end posterior as the convex combination

$$V_i^{\text{post}} \;=\; w(k_i) \cdot V_i^{\text{prior}} \;+\; \big(1 - w(k_i)\big) \cdot \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \hat{v}_m, \qquad w(k_i) \;=\; 0.6 + 0.1\,\min(3,\, k_i), \tag{13}$$

where the weight w ramps deterministically with experience: a novice ($k_i = 0$) moves with a substantial peer contribution ($w = 0.60$), a three-round veteran ($k_i \geq 3$) barely moves ($w = 0.90$). The cap at $k_i = 3$ matches the four-round session structure: after three rounds of experience the agent is at its terminal skepticism for the run. When $\mathcal{M}_i$ is empty—no peer sent a valid claim during the period—Plan I keeps the prior unchanged, $V_i^{\text{post}} = V_i^{\text{prior}}$.

Plan I is the offline baseline to which Plans II and III are compared. Because it consults no external resource, it executes in constant time per period and its output is a pure function of the seeded PRNG plus the received-message set. Two runs with the same (population, seed) pair therefore produce identical trajectories, and the downstream `Market` replay can be reconstructed frame-by-frame from the append-only history arrays without re-running any agent.

C. Plan II: LLM update with explicit utility forms

Plan II replaces equation (13) with a live language model call. Let A_U be the set of utility agents in the current population. At every period boundary, once the messaging round has closed and the trust EMA has updated (Section IV), the engine snapshots each agent's received-message set onto the agent and fires a single `chat/completions` call per agent in parallel. The system prompt is fixed and instructs the model to return a single number with no prose. The user prompt π_i^{II} encodes six facts:

1. the paper constants $(T, \mathbb{E}[d])$ and the current period index t;
2. the agent's risk-preference category $U_i \in \{U_L, U_N, U_A\}$;
3. the *closed-form* expression of the risk-utility functional U_i;
4. the agent's bias mode $b_i \in \{\text{over}, \text{none}, \text{under}\}$;
5. the agent's current mark-to-market wealth $w_i = \text{cash}_i + \text{inventory}_i \cdot p^\star$ with $p^\star = \max(\text{last}, \text{FV}_t)$;

6. the set of peer claims $\{\hat{v}_m\}_{m \in \mathcal{M}_i}$ received during the current period.

The third item is the load-bearing element. Concretely the prompt reads

> You, agent i, are a *risk* trader. Your expected utility over a next-period wealth w takes the closed form
>
> $$U_i(w) \;=\; (w/w_0)^2 \quad / \quad w/w_0 \quad / \quad \sqrt{w/w_0}$$
>
> for risk-loving, risk-neutral, or risk-averse, respectively. Your current mark-to-market wealth is w_i. During the current period your peers broadcast the following private valuations: $\{\hat{v}_m\}$. Market: T periods, expected dividend $\mathbb{E}[d]$, current fundamental value FV_t cents. What is your private subjective valuation per share for next period, in cents?

where *risk* is replaced by the agent's own category. The reply is parsed for the first signed decimal, converted to a float, clamped to a sanity interval, and written to the agent's `llmBeliefs` slot in the engine context. On the next period's first call to `updateBelief`, the slot is consumed in place of equation (13). If the clamp rejects the reply, or the parse fails, or the network errors, the slot is left empty and the agent falls through to Plan I for that period. The next period's LLM call then runs fresh, so a missed update never cascades.

The key feature of Plan II is that the prompt explicitly contains the functional form $(w/w_0)^2$, w/w_0, or $\sqrt{w/w_0}$ for the agent's assigned risk preference. A language model that already knows what a risk-averse agent looks like does not need the formula; a language model that does not know (or that translates "risk-averse" into a qualitatively different curvature) behaves differently when the formula is present. The contrast between Plan II and Plan III measures exactly that.

D. Plan III: LLM update with risk label only

Plan III uses the same period-boundary wiring as Plan II with identical timing, identical clamping, identical fallback, and the same parallel batch of chat completions. The *only* difference is the user prompt: the closed-form expressions are deleted. The prompt states

> You, agent i, are a *risk-loving* / *risk-neutral* / *risk-averse* trader. Your current mark-to-market wealth is w_i. During the current period your peers broadcast the following private valuations: $\{\hat{v}_m\}$. Market: T periods, expected dividend $\mathbb{E}[d]$, current fundamental value FV_t cents. What is your private subjective valuation per share for next period, in cents?

No $U_L/U_N/U_A$ formula is supplied and no reference is made to expected utility; the model must translate the risk label into an implicit curvature internally. Everything else—the mark-to-market wealth, the peer claims, the market constants, the prompt preamble, and the single-number return requirement—is byte-for-byte identical to Plan II.

If Plan II and Plan III produce measurably different distributions over V_i^{post} or over downstream market-quality statistics, the difference is attributable to prompt structure alone. If they agree, the LLM in use has a stable internal representation of the three risk archetypes that is robust to whether the functional form is named in the prompt.

E. Consume-and-cache on the period boundary

The LLM call in Plan II and Plan III is fired *after* the period's messaging round closes but *before* the next period's first decision tick, so that the prompt can include the peer claims observed during the period that just ended. Concretely the engine:

1. copies each agent's `receivedMsgs` array into a snapshot field, so the downstream call reads a stable message set;

2. fires one `chat/completions` call per agent in a single `Promise.all` batch, so network latency is one round trip rather than $|A_U|$ round trips;

3. merges each successful reply into a context-level `llmBeliefs` map keyed by agent id;

4. on the next period's first call to `updateBelief`, consumes the entry in place of equation (13) and clears the map slot so a second call in the same period reverts to Plan I.

This *consume-and-cache* pattern keeps the tick loop synchronous and non-blocking: agents continue to quote and trade at the usual tick cadence while the LLM batch runs in the background, and the result is applied on the very next period boundary. If the call is still in flight when the next boundary arrives, the `llmBeliefs` slot is empty and the agent runs Plan I for that period, so the engine never stalls waiting for a network response.

F. Determinism and reproducibility

Plan I is fully deterministic under the seeded mulberry32 PRNG: two runs with the same (population, seed) pair produce byte-identical trace, snapshot, and event logs. Plan II and Plan III introduce one source of non-determinism—the LLM reply—and the engine handles it by keeping the fallback path closed-form. If the same seed is run twice under Plan II, the two runs will in general disagree wherever the language model's reply differs between calls, but both runs will produce valid simulations and both will respect the clamp and the parse. Reporting therefore always pairs a plan label with a seed when comparing statistics across runs.

A missing reply, a parse failure, or an HTTP error silently drops the update for that agent in that period, and the agent falls through to Plan I. This preserves the invariant that every run can be reproduced without network access—Plan II with zero successful calls is definitionally identical to Plan I—so the simulator can be used as a pedagogical or offline replay tool even when the API endpoint is unreachable.

VI. Preliminary Results

Because the simulator is designed for interactive experimentation rather than a single calibrated empirical test, this section reports preliminary findings from matched runs across the three belief-update plans of Section V. Unless otherwise noted, every comparison holds the market structure, the population mix, the endowment draw, and the engine seed fixed, and varies only the plan label. Plan I runs are rolled from the seeded PRNG with no network access; Plan II and Plan III runs consume one chat completion per utility agent per period-boundary. A Plan II or Plan III run with zero successful LLM calls is definitionally identical to the matching Plan I run, so the fallback path is treated as the null hypothesis against which the LLM channels are compared.

A. Plan I baseline: experience-weighted blending

Under Plan I the utility agents' period-boundary posterior follows the closed form of equation (13): a convex combination of the Lopez-Lira prior and the mean foreign peer claim, with weight $w(k_i) = 0.6 + 0.1\,\min(3, k_i)$. In round 1 every utility agent has $k_i = 0$, so $w = 0.60$ and the mean peer claim contributes 40% of the posterior. Amplitude A typically falls in the 0.20–0.40 range, ND in 0.10–0.25, and Haessel's R^2 rises from near zero in the first few periods to roughly 0.40–0.80 by round 3 as the prior weight ramps toward 0.90 and the agents' own FV-anchored beliefs dominate the blend. The bubble–crash signature is present but substantially attenuated compared with the pure Lopez-Lira $w = 0.60$ setting used in Lopez-Lira (2025): the experience ramp plus the F/T/R background supplies enough rational anchor to contain the largest excursions by round 4.

B. Plan II: LLM update with explicit utility forms

Under Plan II each period boundary fires a parallel batch of LLM calls, one per utility agent. The prompt includes the closed-form $U_L/U_N/U_A$ expression for the agent's risk type, the agent's mark-to-market wealth, and the set of peer claims received during the just-ended period. The returned number is clamped and written to next period's prior slot. Three preliminary observations are clear from a handful of runs.

Risk-typed stratification.. The LLM's replies are heterogeneous across risk archetypes in the expected direction. Risk-loving, persistently-optimistic agents receive period-end posteriors above the per-period FV_t; risk-averse, persistently-pessimistic agents receive posteriors visibly below FV_t; risk-neutral unbiased agents cluster near FV_t. The simulator reports the full posterior vector in its subjective-valuation panel, so the reader can verify that the declared risk preferences map into observable valuation differences without re-parsing the logs.

Between-period smoothing.. Because the LLM update is fired once per period (not once per tick), the posterior trajectory is a piecewise-constant staircase within a period and a discrete jump at the boundary. The first-decide tick of each new period is therefore visibly stratified by the posterior that came out of the just-completed LLM batch, but the effect is absorbed within a few ticks as the expected-utility score consumes the new prior and the agents resume quoting under (11).

Amplitude relative to Plan I.. Matched-seed Plan II runs typically produce slightly higher amplitude and slightly lower R^2 than Plan I. The LLM posterior is less tightly anchored to the prior than the $w = 0.6$–0.9 algorithmic blend, so the resulting utility-agent orders are more dispersed around FV_t. Whether this dispersion reflects a genuine difference in the model's internal expected-utility reasoning or a noise term in the LLM's reply distribution is a question the full-replication study would answer by running many seeds per condition.

C. Plan III: LLM update with risk label only

Under Plan III the same prompt goes out but the closed-form $U_L/U_N/U_A$ expressions are stripped from the user message. The LLM is told only that the agent is, e.g., risk-averse, and must translate that label into an implicit curvature internally. Matched-seed comparisons against Plan II answer a narrow empirical question: does naming the functional form shift the returned valuations in a way that moves downstream market statistics?

In preliminary runs the Plan II / Plan III gap is small. The posterior distributions overlap heavily across the three risk archetypes; per-period mean prices track within a few cents of each other for most seeds; and the bubble statistics A, ND, and R^2 are statistically indistinguishable within a handful of matched-seed replications. This is consistent with an LLM whose internal representation of the three risk labels already encodes the curvature implied by the functional forms—naming them in the prompt provides redundant information rather than new signal. A full study with many seeds per condition would establish whether the residual Plan II / Plan III gap is statistical noise or a genuine (small) prompt-sensitivity effect.

D. Market-quality statistics at a glance

Table 1 summarises the canonical bubble statistics under the three plans in qualitative—not yet calibrated—form. Each column reports a range rather than a point estimate; a replication study would tighten each range by running many seeds per plan and would add a permutation test for between-plan equality.

Table 1: Market-quality statistics across the three plans, qualitative ranges

	Plan I	Plan II	Plan III
Haessel R^2	0.40–0.80	0.30–0.75	0.30–0.75
Normalised dev. ND	0.10–0.25	0.10–0.30	0.10–0.30
Amplitude A	0.20–0.40	0.25–0.45	0.25–0.45
Turnover TO	0.5–1.0	0.5–1.0	0.5–1.0
Allocative eff. AE	0.7–0.95	0.6–0.90	0.6–0.90

The pattern in Table 1 is consistent with three hypotheses: first, that Plan I's experience-weighted blend produces the cleanest convergence to the fundamental because it is the only plan that ramps the prior weight monotonically toward one across the session; second, that Plans II and III introduce additional dispersion through the LLM reply distribution without producing a qualitatively different market regime; and third, that the gap between Plan II and Plan III is within the resolution of the qualitative table, implying that the language model's internal representation of risk labels is already robust to whether a closed-form curvature is named in the prompt. A properly replicated experiment with dozens of seeds per cell—and formal permutation tests—is required to turn these hypotheses into findings; the testbed is built to support that experiment, not to substitute for it.

VII. Conclusion

This paper documents a browser-based replication of the experimental asset market of Dufwenberg et al. (2005), populated with the expected-utility messaging agent of Lopez-Lira (2025), and compares three distinct plans for the utility agent's per-period belief update: an algorithmic experience-weighted blend (*Plan I*), a language-model call whose prompt contains the closed-form risk-utility expression for the agent's type (*Plan II*), and an otherwise-identical language-model call whose prompt names only the risk-preference category (*Plan III*). The three plans hold the market substrate, the agent population, the starting prior, the expected-utility action choice, the order-matching step, the dividend credit, and the market-quality instruments strictly fixed, so any difference in observed market statistics is attributable to the belief-update channel alone. The seeded mulberry32 generator ensures that Plan I replays are byte-identical for a given (population, seed) pair, and Plan II and Plan III silently degrade to Plan I for any agent whose LLM call fails, so every run can be reproduced without network access.

The contribution is threefold. First, the testbed constitutes a clean vanilla-JavaScript implementation of the DLM 2005 market with Lopez-Lira utility agents on top, specified in enough detail (Sections III–IV) that a reader can reconstruct the model from the paper alone. Second, factoring the utility agent's belief update into three distinct plans provides a narrow but interpretable lever for comparing algorithmic decision-making against language-model decision-making at the exact point in the pipeline where the difference can be measured: the posterior valuation, a single number per agent per period. Third, the prompt-structure contrast between Plan II and Plan III isolates a specific question about language models: does handing the model a closed-form utility function change its returned valuations in a way that propagates into downstream market dynamics?

Three caveats shape the preliminary findings. First, the ranges in Table 1 are illustrative, not calibrated: a full empirical test requires many replications per cell, a pooled treatment of the LLM-reply variance, and a formal permutation test between plans. Second, the LLM update fires once per period rather than once per tick, bounding the within-period effect of the plan on quoting behaviour; a faster-clocked variant would be a natural extension but would impose a much heavier network load. Third, the specific language model used for Plan II and Plan III affects the within-plan distribution of posteriors; a full study would fix a model version and report per-model comparisons separately.

The testbed is free of build steps and of third-party dependencies, and ships with the architecture diagram, glossary, and slide deck used in this paper. Its "Experiment settings" panel exposes every numerical constant that shapes a run—the population mix, the risk-preference share, the passive fill probability, the trust learning rate, the bias amplitude, and the valuation noise—and the seeded PRNG makes every edit to a constant observable as a difference against a matched baseline. The plan switcher opens several follow-ups: whether Plan II and Plan III agree across model families, whether a Plan IV that replaces the mean-peer term with a trust-weighted peer term matches the LLM plans, and whether the posterior trajectory under Plan II or Plan III predicts the holder of the asset at session end any better than the Plan I baseline.

References

Dufwenberg, Martin, Tobias Lindqvist, and Evan Moore, "Bubbles and Experience: An Experiment," *American Economic Review*, 2005, *95* (5), 1731–1737.

Gode, Dhananjay K. and Shyam Sunder, "Allocative Efficiency of Markets with Zero-Intelligence Traders: Market as a Partial Substitute for Individual Rationality," *Journal of Political Economy*, 1993, *101* (1), 119–137.

Haessel, Walter, "Measuring Goodness of Fit in Linear and Nonlinear Models," *Southern Economic Journal*, 1978, *44* (3), 648–652.

Long, J. Bradford De, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann, "Positive Feedback Investment Strategies and Destabilizing Rational Speculation," *Journal of Finance*, 1990, *45* (2), 379–395.

Lopez-Lira, Alejandro, "Can Large Language Models Trade? Testing Financial Theories with LLM Agents in Market Simulations," 2025. Working paper, University of Florida.

Smith, Vernon L., Gerry L. Suchanek, and Arlington W. Williams, "Bubbles, Crashes, and Endogenous Expectations in Experimental Spot Asset Markets," *Econometrica*, 1988, *56* (5), 1119–1151.

Stöckl, Thomas, Jurgen Huber, and Michael Kirchler, "Bubble Measures in Experimental Asset Markets," *Experimental Economics*, 2010, *13* (3), 284–298.